

Final Exam
Logic and Sets for AI

This exam has 3 pages and 6 exercises with a total of 66 points. The result will be computed as $1 + \left(\frac{\text{total}}{66} \cdot 9\right)$.

Note that in this course, 0 is considered a natural number. If you are, in any exercise, unsure about the definition of some term/operator/... state your assumed definition at the beginning of the exercise.

*Please put **your name on every sheet** you hand in, and write as clearly as possible. Submissions (even parts) without a name might **not be considered**. If we cannot read a statement it will be considered **incorrect**.*

Good luck!

1. Equivalence classes for $\equiv$ (9 points points)

Group the following twelve formulas into their semantic equivalence classes:

- | | | |
|--|--|---|
| (1) p | (2) $p \rightarrow (\neg q \wedge \neg p)$ | (3) $\neg p \rightarrow (\neg q \wedge \neg p)$ |
| (4) $\neg p \rightarrow (q \wedge p)$ | (5) $p \rightarrow (\neg q \wedge p)$ | (6) $\neg p \rightarrow (\neg q \wedge p)$ |
| (7) $p \wedge (q \rightarrow \neg p)$ | (8) $\neg p \wedge (q \rightarrow \neg p)$ | (9) $p \wedge (q \rightarrow \neg p)$ |
| (10) $\neg p \wedge (q \rightarrow p)$ | (11) $p \wedge (\neg q \rightarrow p)$ | (12) $\neg p$ |

2. OBDDs/Logic circuits (5+4+5 points)

- (a) Draw the binary decision tree for the following formula. Use the order p, q, r for the variables. Apply the reduction algorithm to the tree until you've arrived at a reduced OBDD. Make sure to name all the steps.

$$(p \rightarrow q) \vee (q \oplus p)$$

- (b) Draw the circuit diagram for the formula.
- (c) Draw the OBDDs for both disjuncts of the equation separately, denoted as O_1 and O_2 . Use them to construct the OBDD for $O_1 \wedge O_2$.

3. Predicate Logic (2+8 points)

- (a) Using the following predicates and variables represent the sentence *"If the Earth orbits the Sun, it is a planet."*

$O(x,y) := x$ orbits around y

$P(x) := x$ is a planet

$e :=$ The Earth

$s :=$ The Sun

- (b) Add one predicate to the above. Use all three predicates and any number of quantifiers to represent the following sentences:

(i) *"Some planets have a moon orbiting around them."*

(ii) *"Everything that orbits around the sun is either a planet or a moon."*

Note that a planet cannot be a moon.

4. Equivalence Relations (3+6 points)

The set $\mathbb{Z} \setminus \{0\}$ (non-zero integers) can be partitioned into the following four sets: the set of positive even, positive odd, negative even, and negative odd numbers. Let R be the equivalence relation in $\mathbb{Z} \setminus \{0\}$ that has these four sets as its equivalence classes.

- (a) Give a complete system of representatives for R .
- (b) Consider the relation S on $\mathbb{N}$ that is defined by

$$xSy \iff \text{sum of digits in } x = \text{sum of digits in } y$$

.

For example, $14 S 212$, as the sum of digits in both numbers is 5. Show that S is an equivalence relation.

5. Functions (6+3+5 points)

Consider the functions $sqr : \mathbb{R} \rightarrow \mathbb{R}$, $add_2 : \mathbb{R} \rightarrow \mathbb{R}$, and $mul_2 : \mathbb{R} \rightarrow \mathbb{R}$, which are defined as follows:

$$sqr(x) := x^2, \quad add_2(x) := x + 2, \quad mul_2(x) := x * 2$$

- (a) Give a description (i.e., formula for $f(x)$) for the function f defined below and provide its domain of definition and range.

$$f := mul_2^{-1} \circ sqr^{-1} \circ add_2$$

- (b) Give the inverse of f as a composition of sqr , add_2 , and mul_2 .
- (c) Now consider the function $abs : \mathbb{R} \rightarrow \mathbb{R}$, where $abs(x) = |x|$ denotes the absolute value of x . Additionally, the function g is defined as $g(x) = abs \circ mul_2$.

Is the function g total, injective, and/or surjective?

6. Induction (2+8 points)

We claim that the following statement is true for all natural numbers $n \geq 1$.

$$\sum_{k=1}^n k * (k + 1) = \frac{n * (n + 1) * (n + 2)}{3}$$

- (a) Verify by explicit computation that the claim is true for $n = 1$ and $n = 2$.
- (b) Prove by mathematical induction that the statement holds true for all natural numbers $n \geq 1$.

End of the exam!